An Empirical Analysis of ChatGPT's Impact on Developer Productivity Using Italy's ChatGPT Ban as a Natural Experiment

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1 Introduction

Substantial anecdotal evidence suggests that AI-assisted coding tools such as ChatGPT [9] have significantly increased developer productivity. However, recent empirical research has shown mixed results [4, 10, 11]. While some studies find notable efficiency gains from integrating generative AI into software development workflows—for instance, faster code completion and reduced debugging time—other work points to limited or highly context-dependent effects, with improvements varying by task complexity, developer experience, and integration depth.

Despite this growing literature, much of the existing evidence is correlational or based on controlled experiments detached from real-world settings. We still know little about how sudden changes in access to such tools affect developers in practice. The short-lived ChatGPT ban in Italy in early 2023 [6] offers a rare quasi-experimental opportunity to observe developers' behavioral responses to an abrupt disruption in access to a key AI tool.

By analyzing GitHub repositories, Kreitmeir and Raschky [8] documented a 50% decline in release events in Italian open source projects during the first days of the ban. However, the effects on the granular, day-to-day development process remain unquantified. It remains unclear whether developers wrote less code or it took them longer to complete individual features.

Building on Kreitmeir and Raschky's insights, we investigate additional effects the ban has had on developers by focusing on code and feature-level metrics. While release events capture aggregate project milestones, they provide limited insight into the underlying development dynamics. A release might include multiple new features, and might therefore consist of the work accumulated over a longer period of time, making it more difficult to show immediate impacts. Code-level metrics—specifically lines of code added and deleted per developer per day—offer a direct, granular measure of coding activity that reveals whether the sudden loss of ChatGPT access led to immediate changes at the level of individual code production. Pull requests offer another perspective on developer productivity. As they usually

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encompass the code changes required to implement a new feature or resolve a bug, their frequency serves as a proxy for feature-level output.

To investigate how the ChatGPT ban affected developers' coding activity, we address the following research questions:

- **RQ1**: Did the ChatGPT ban cause a reduction in the daily volume of code contributed by developers in Italy, as measured through lines of code added and deleted per user per day?
- **RQ2**: Did the ChatGPT ban lead to a decrease in the number of pull requests opened by developers in Italy, indicating a reduction in feature-level output?

These research questions allow us to examine the immediate behavioral response, specifically coding activity and feature-level output, to the ban. By focusing on granular metrics, we aim to provide a more comprehensive understanding of how the disruption in AI tool access affected the day-to-day development process.

2 Related Work

2.1 AI-Assisted Coding Tools and Developer Productivity

A growing body of empirical research has investigated the impact of AI-assisted coding tools on developer productivity. Early studies on GitHub Copilot found significant productivity gains, with developers completing tasks faster when using AI code completion [11]. More recent large-scale field experiments have provided mixed evidence: Cui et al. [4] conducted randomized controlled trials across three companies and found a 26% increase in completed tasks among developers using AI tools, with less experienced developers showing greater gains. However, Paradis et al. [10] reported more context-dependent effects in enterprise settings, suggesting that productivity improvements vary substantially by task complexity and developer experience. Becker et al. [3] examined open-source developers and found nuanced effects that depend on the type of development work being performed.

While these studies provide valuable insights, they primarily rely on controlled experiments or observational data that may not capture real-world behavioral responses to sudden changes in tool availability. Most existing work focuses on productivity gains during tool adoption, leaving open the question of how developers adapt when access to these tools is abruptly disrupted.

2.2 The ChatGPT Ban as a Natural Experiment

Kreitmeir and Raschky [8] were the first to leverage Italy's ChatGPT ban as a quasi-experimental setting to study the real-world impact of AI tool access. Using a difference-in-differences design, they analyzed GitHub release events and documented a 50% decline in releases from Italian open-source projects during the first days of the ban. This work established the ban as a valuable natural experiment and demonstrated the feasibility of using GitHub activity data to study developer behavior at scale.

However, their analysis focused on aggregate project-level outcomes (release events), which capture only high-level project milestones. A release may bundle multiple features and code changes, making it less sensitive to immediate shifts in day-to-day coding activity. Additionally, releases might be influenced by factors beyond individual developer productivity, such as project management practices and team coordination. The granular, day-to-day development process—including actual code production indicators—remains unexamined. Our work extends this foundation by investigating code-level and issue-level metrics that provide more direct measures of developer activity.

2.3 Measuring Developer Productivity

Prior research has demonstrated that developer productivity is influenced by multiple factors and, therefore, cannot be accurately captured by a single metric [5]. Moreover, measuring output quantity—such as the number of lines of code a developer adds—can encourage behavior aimed at optimizing the metric rather than achieving meaningful progress [13]. However, when working with historical data, alternative metrics are often unavailable. In such cases, and particularly when developers are unaware that their output is being measured, the risk of such gaming is relatively low. Accordingly, in this study, we use activity measures such as lines of code added and deleted, as well as pull requests opened, as proxies for developer productivity, while acknowledging the inherent limitations of this approach.

3 Methodology

3.1 Data

The empirical analysis uses historical GitHub activity data from users in Italy, Austria, and France for the weeks surrounding the early April 2023 ban. We leverage the GH Archive dataset, which provides a comprehensive record of public GitHub events [1]. Since GH Archive data does not include developers' location information, we infer user countries based on their public profiles, following the methods from prior research [8].

3.2 Control Group Selection

We follow Kreitmeir and Raschky [8] in selecting Austria and France as the control group for our difference-in-differences analysis. This choice is motivated by several factors that strengthen the validity of the parallel trends' assumption, which is critical for causal inference in DiD designs [2]. First, Austria and France share cultural and linguistic similarities with Italy, being neighboring European countries with comparable levels of economic development and technology adoption. Second, their geographic proximity means they are likely subject to similar regional trends in software development practices, AI tool adoption, and broader technological developments. Third, these countries were not affected by the ChatGPT ban, making them suitable counterfactuals for what would have happened in Italy in the absence of the ban. This selection helps ensure that any observed differences between Italy and the control countries during the ban period are attributable to the ban itself rather than to pre-existing differences in development trends.

3.3 Research Design

We employ a **Difference-in-Differences (DiD)** quasi-experimental design to ensure causal inference, replicating the framework from the original paper. The analysis compares the change in developer metrics in Italy (treatment group) against those in Austria and France (control group).

The ban was implemented on April 1, 2023. Our analysis focuses on a time window spanning four weeks before the ban (March 4–31, 2023) and two weeks after the ban (April 1–14, 2023). This time window allows us to capture both the immediate impact of the ban and potential adaptation effects. Additionally, following Kreitmeir and Raschky [8], we examine the first four working days after the ban (April 3–6, 2023) separately to assess the immediate impact before circumvention behavior (such as VPN usage, as documented in the appendix) may have mitigated the ban's effects over time. This design allows us to address the research questions (RQ1 and RQ2) by examining whether the ban caused statistically significant changes in code volume and bug-related issue creation.

In contrast to the original study—which measured aggregate productivity through project releases—our analysis zooms in on code-level and issue-level metrics that capture the day-to-day development process. The main dependent variables are:

- **Code Volume**: The number of lines of code (LOC) added and deleted per developer per day. This provides a granular measure of active coding behavior.
- **Pull Requests Opened**: The number of pull requests opened per developer per day, serving as a proxy for feature-level output.

All variables will be standardized at the developer-day level to account for individual differences in coding activity. We include user and date fixed effects in our regression specifications to control for unobserved heterogeneity across developers and time-invariant day-specific factors that may confound the treatment effect. User fixed effects account for individual developer characteristics (e.g., skill level, coding style, project preferences) that remain constant over time, while date fixed effects capture common temporal shocks affecting all developers on a given day (e.g., holidays, major technology events, or platform-wide changes). We also include an interaction term between Italy and a linear time trend (Italy × Trend) to account for potential differential pre-treatment trends between Italy and the control countries, though its coefficient is not reported in the results table as it is not of primary interest.

To explore whether the ban's effects varied across different types of developers, we conduct a heterogeneity analysis by stratifying developers into activity groups based on pre-ban commit volume tertiles. This analysis allows us to examine whether developers with different baseline activity levels responded differently to the ban, which could provide insights into how AI tool usage patterns vary across developer types.

4 Results

4.1 Code Volume Analysis

We begin by examining the impact of the ChatGPT ban on developers' coding activity, as measured by lines of code (LOC) added and deleted per developer per day. Table 1 presents the difference-in-differences regression results for both the two-week post-ban period and the first four working days after the ban implementation.

Table 1. Difference-in-Differences Results: LOC Added / Deleted (Absorbed Fixed Effects)

	Log Additions		**Log Deletions**	
	2 Weeks	4 Weekdays	2 Weeks	4 Weekdays
Italy × Post	0.0177	0.0542	0.0136	0.0326
	(0.0486)	(0.0527)	(0.0489)	(0.0524)
Observations	224, 776	185, 999	224, 776	185, 999
R^2	0.428	0.449	0.425	0.444
User FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

Note: *** p<0.01, ** p<0.05, * p<0.1

The coefficient on "Italy × Post-Ban" represents the treatment effect of the ban on coding activity in Italy relative to the control countries (Austria and France). After controlling for user and date fixed effects, we find no statistically significant effects on either lines of code added or deleted in either time period. The coefficients are small in magnitude and not statistically significant at conventional levels. For log additions, the treatment effects are 0.0177 (SE: 0.0486) over the two-week period and 0.0542 (SE: 0.0527) over the four-weekday period. For log deletions, the corresponding effects are 0.0136 (SE: 0.0489) and 0.0326 (SE: 0.0524), respectively.

These null results are likely explained by widespread ban circumvention through VPN usage and other methods. As documented in Appendix A, VPN searches in Italy increased by 28.6 points (on a 0–100 scale) more than in control countries following the ban implementation, an effect that is statistically significant at the 1% level. This substantial increase in circumvention behavior suggests that many developers quickly regained access to ChatGPT despite the ban, effectively nullifying the treatment. If developers circumvented the ban and continued using ChatGPT, we would expect to observe no reduction in coding activity—which is precisely what we find. The inclusion of fixed effects is important for causal identification, as it controls for time-invariant developer characteristics and common day-specific shocks that could otherwise confound the treatment effect.

These results address RQ1, indicating that we do not find statistically significant evidence that the ChatGPT ban caused a reduction in the daily volume of code contributed by developers in Italy when controlling for developer and temporal heterogeneity. The null findings are consistent with effective circumvention of the ban rather than an absence of AI tool effects on coding activity.

4.1.1 Heterogeneity Analysis. To explore whether the ban's effects varied across different types of developers, we conduct a heterogeneity analysis by developer activity level. We stratify developers into three groups (low, mid, and high activity) based on pre-ban commit volume tertiles. Table 2 presents the difference-in-differences estimates separately for each activity group.

Table 2. Heterogeneity Analysis by Developer Activity Level

	Log Additions			**Log Deletions**		
	Low	Mid	High	Low	Mid	High
Italy × Post	0.0238	0.0741	0.0016	-0.0908	-0.0196	0.0192
	(0.2172)	(0.1085)	(0.0560)	(0.2145)	(0.1068)	(0.0565)
Observations	17,424	42,697	157,932	17,424	42,697	157,932
User FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Activity groups defined by pre-ban commit volume tertiles. Standard errors clustered by user in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

The results show no statistically significant heterogeneity in the ban's impact across developer activity levels. For log additions, the treatment effects are 0.0238 (SE: 0.2172) for low-activity developers, 0.0741 (SE: 0.1085) for mid-activity developers, and 0.0016 (SE: 0.0560) for high-activity developers. For log deletions, the corresponding effects are -0.0908 (SE: 0.2145), -0.0196 (SE: 0.1068), and 0.0192 (SE: 0.0565) for low, mid, and high-activity developers, respectively. None of these coefficients are statistically significant at conventional levels, indicating that we do not find evidence of differential effects of the ban across developer activity groups. The null results across all activity levels are consistent with the main findings and suggest that the ban did not have statistically detectable impacts on coding activity regardless of developers' baseline activity levels.

4.1.2 Ban Lift Analysis. To further examine the causal relationship, we analyze the period following the ban's lift on April 28, 2023. Table 3 presents difference-in-differences estimates comparing coding activity before and after the ban was lifted. The coefficient on "Italy × Post-Lift" represents whether developers in Italy experienced a rebound in coding activity relative to the control countries once ChatGPT access was restored.

Table 3. Difference-in-Differences Results: Post-Ban Lift (Absorbed Fixed Effects)

	Log Additions		Log Deletions	
	2 Weeks	4 Weekdays	2 Weeks	4 Weekdays
Italy × Post-Lift	-0.0321	-0.0158	-0.0044	-0.0585
	(0.0494)	(0.0541)	(0.0502)	(0.0547)
Observations	209142	169574	209142	169574
R^2	0.436	0.457	0.430	0.451
User FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

Note: *** p<0.01, ** p<0.05, * p<0.1

We find no statistically significant effects in any of the specifications. The coefficients on "Italy × Post-Lift" are small in magnitude and not statistically significant for both LOC added and deleted, across both the two-week and four-weekday time windows.

The absence of a rebound effect is consistent with our main findings and provides the strongest evidence for our interpretation: **if there was no initial dip in coding activity due to circumvention, there can be no rebound when the ban is lifted**. As documented in Appendix A, VPN searches in Italy increased by 28.6 points immediately following the ban implementation, indicating widespread circumvention that allowed developers to maintain access to ChatGPT throughout the ban period. By the time the ban was officially lifted on April 28, many developers had already circumvented the restriction weeks earlier, meaning the lift had no meaningful impact on their access. The null results in both the ban period and the ban lift period are therefore mutually reinforcing: they both point to effective circumvention as the explanation for why we observe no treatment effects on coding activity.

4.2 Pull Requests Opened Analysis

To address RQ2, we examine the impact of the ChatGPT ban on the number of pull requests opened by developers in Italy. Table 4 presents the difference-in-differences regression results for pull requests opened per developer per day, using the same time windows as in the code volume analysis.

Table 4. Difference-in-Differences Results: Pull Requests Opened (Absorbed Fixed Effects)

	Log Pull Requests Opened	
	2 Weeks	4 Weekdays
Italy × Post	0.0175	0.0306*
	(0.0162)	(0.0181)
Observations	47, 943	39, 794
R^2	0.387	0.402
User FE	Yes	Yes
Date FE	Yes	Yes

Note: *** p<0.01, ** p<0.05, * p<0.1

"Italy × Post-Ban" captures the treatment effect of the ban on pull request activity in Italy relative to the control countries (Austria and France). The results show no statistically significant effects on pull requests opened in the two-weeks period. We observe a treatment effect of 0.0326 (SE: 0.0524) for the four-weekday period, indicating a slight increase of pull requests opened. This effect is statistically significant at the 10% level (p=0.0897), suggesting a weak positive impact of the ban on pull request activity during this shorter time frame. However, given the small effect size and limited statistical significance, this result may reflect random variation rather than a true treatment effect.

5 Discussion

5.1 The Resilience of Developer Workflows

Contrary to expectations derived from recent field experiments, which documented significant productivity gains from AI adoption [4, 11], our analysis revealed no statistically significant reduction in coding volume following the ChatGPT ban. We argue that this null result should not be interpreted as evidence that Generative AI lacks utility, but rather as evidence of the high *resilience* of developer workflows against external disruptions.

This interpretation is strongly supported by our analysis of search trends, which shows a dramatic, immediate spike in VPN search interest in Italy—reaching peak popularity (index 100) on the very day the ban was implemented (see Figure 1). While Google Trends data aggregates interest across the general population, software developers possess the specific *technological fluency* required to rapidly adopt circumvention tools.

Viewed through the lens of organizational psychology, this behavior represents an immediate shift to "Shadow IT," where the perceived cognitive value of the AI tool outweighed the friction of non-compliance. Consequently, the "treatment" in this natural experiment was likely porous; the stability of the Lines of Code (LOC) metric suggests that developers successfully maintained their AI-augmented workflows by bypassing the geographic restriction, effectively treating the ban as a technical hurdle rather than a hard constraint.

Another reason for the resilience of developer workflows could be the availability of alternative AI tools. Although the ChatGPT website was blocked, access to the OpenAI API remained possible [12]. Access to AI tools integrated into development environments, such as GitHub Copilot, was also unaffected.

These possibilities to circumvent the ban show that developers might have found alternative ways to access AI assistance, thereby maintaining their productivity levels. Also, it should be considered that ChatGPT was still relatively new at the time of the ban, which might have limited its integration into developers' workflows. As developers become more accustomed to using AI tools, future disruptions could have more pronounced effects.

6 Conclusion

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A Ban Circumvention Analysis

This appendix presents the difference-in-differences analysis of search volume for circumvention tools—specifically VPN services and Tor Browser—in Italy (treatment group) compared to Austria and Germany (control groups) before and after the ban implementation on April 1st, 2023. The data was obtained from Google Trends [7], which provides normalized search interest values representing the share of web searches for a given term relative to all searches on Google over time. The values are normalized on a scale of 0–100, where 100 represents peak popularity for the term during the selected time period.

Figure 1 shows the VPN search volume over time for Italy, Austria, and Germany. The vertical dashed line indicates the ban start date (April 1st, 2023). Prior to the ban, Italy showed consistently lower baseline search interest in VPNs compared to the control countries, with average values around 21 points versus 33–35 points for Austria and Germany. Following the ban implementation, Italy experienced a dramatic spike in VPN search interest, reaching a peak of 100 (the maximum normalized value) on April 1st, and maintaining elevated levels throughout the post-ban period with an average of 51 points.

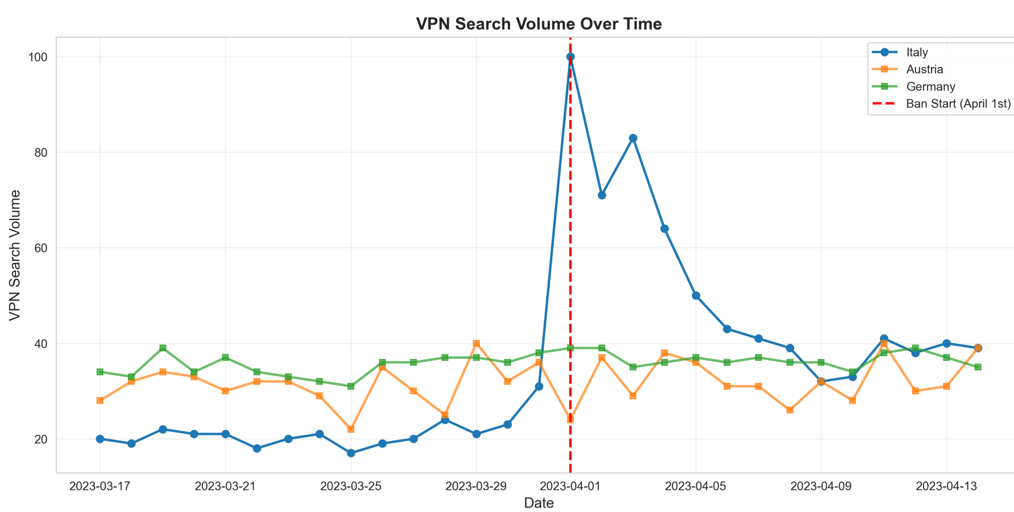

Fig. 1. VPN Search Volume Over Time

Figure 2 compares the average VPN search volume before and after the ban implementation across the three countries. The visualization clearly demonstrates the substantial increase in VPN search interest in Italy following the ban, while the control countries (Austria and Germany) showed minimal changes, consistent with the parallel trends assumption required for difference-in-differences estimation.

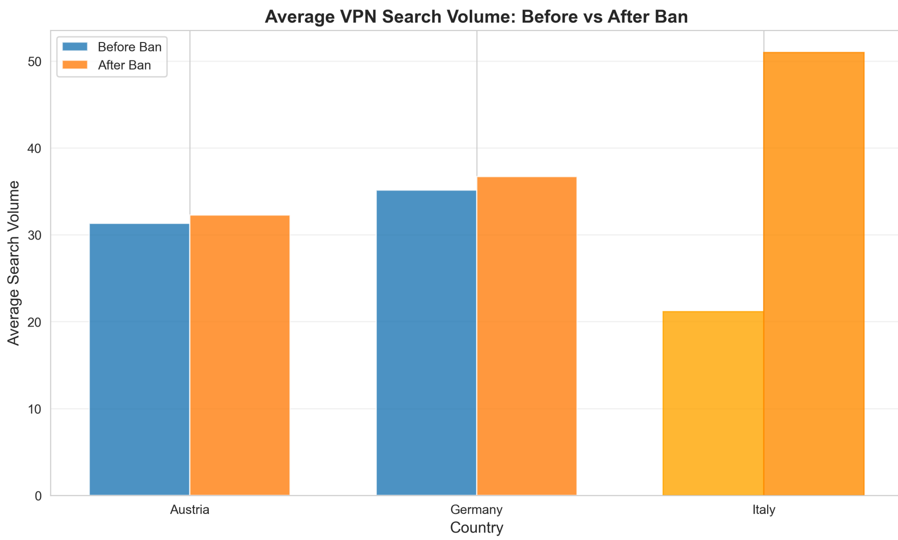

Fig. 2. Average VPN Search Volume: Before vs After Ban

Table 5 presents the difference-in-differences regression results for both VPN and Tor Browser searches. The dependent variable in each model is search volume (normalized Google Trends index). The coefficient on "Italy × Post-Ban" represents the treatment effect for each circumvention tool. For VPN searches, the coefficient indicates that searches in Italy increased by 28.6 points (on the 0–100 scale) more than in the control countries after the ban implementation, an effect that is statistically significant at the 1% level ($p < 0.001$). For Tor Browser searches, the treatment effect is 20.2 points, statistically significant at the 5% level ($p < 0.05$). These results suggest a substantial increase in interest in both VPN services and Tor Browser as means of circumventing the ban. The magnitude of the VPN effect represents more than a doubling of baseline search interest in Italy, consistent with users seeking tools to bypass geographic restrictions.

Table 5. Difference-in-Differences Regression Results

	Dependent variable: search_volume	
	VPN	Tor Browser
Italy	-12.100***	-10.167
	(2.823)	(6.221)
Post-Ban	1.267	-0.710
	(2.346)	(5.169)
Italy × Post-Ban	28.600***	20.167**
	(4.063)	(8.953)
Constant	33.233***	17.567***
	(1.630)	(3.592)
Observations	87	87
R^2	0.497	0.079
Adjusted R^2	0.479	0.046
Residual Std. Error	8.928 (df=83)	19.672 (df=83)
F Statistic	27.344*** (df=3; 83)	2.371* (df=3; 83)
Note:		*p<0.1; **p<0.05; ***p<0.01